BGC Trust University Bangladesh

Department of Computer Science and Engineering

A Thesis Proposal on

CrossMamba-Agri:

Cross-Attention Vision Mamba for Joint Pest and Disease Classification

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Contents

1 Introduction and Background

Global agricultural production consistently loses an estimated 20–40% of potential yields to pests and pathogens, directly threatening food security and the economic stability of smallholder farmers in developing nations. Accurate, early identification of both the pathogen responsible for a disease and the entomological pest responsible for infestation is critical; however, farmers typically lack the expertise to distinguish between the two, leading to the misapplication of agrochemicals. Deep learning approaches have successfully automated visual diagnostics, but they remain highly fragmented. CNNs achieve near-perfect accuracy on lab-controlled datasets like PlantVillage but suffer severe degradation when exposed to field environments. Vision Transformers mitigate this by capturing global contextual relationships through self-attention, yet their $O(N^2)$ computational complexity restricts their viability on resource-constrained edge devices such as smartphones and Unmanned Aerial Vehicles (UAVs). Recently, State Space Models (SSMs), particularly the Mamba architecture, have emerged as a compelling alternative, offering linear-time sequence modeling that captures long-range spatial dependencies without the prohibitive memory overhead of self-attention. Despite the rapid adoption of SSMs in isolated agricultural tasks, current frameworks remain single-purpose, failing to address the fundamental need for a unified diagnostic tool capable of processing both pest and disease imagery through a shared representation space.

2 Problem Statement

The development of field-ready agricultural diagnostics is currently hindered by three interacting limitations in the state-of-the-art:

1. **Architectural Separation of Diagnostic Tasks:** Existing models treat pest and disease detection as mutually exclusive domains. A deployed tool that only addresses one class of agricultural threat forces users to pre-diagnose the nature of the issue, severely limiting practical utility in real-world farming contexts.
2. **Computational Bottlenecks in Global Feature Extraction:** While ViTs capture the necessary global context for subtle symptom identification, their quadratic attention costs preclude high-resolution inference on edge hardware.
3. **Vulnerability to Complex Field Backgrounds:** Agricultural imagery is characterized by high background interference (soil, varying illumination, overlapping foliage). Generic visual backbones, including baseline Vision Mamba models, lack task-specific mechanisms to suppress these non-informative high-frequency signals, leading to degraded localization of disease lesions and camouflaged pests.

3 Aim and Objectives

3.1 Aim

To formulate, implement, and validate CrossMamba-Agri—a computationally efficient, dual-purpose Vision Mamba architecture that leverages shared representations, background suppression, and task-conditioned cross-attention to classify plant diseases and insect pests jointly.

3.2 Specific Objectives

1. To design a Background Suppression Module (BSM) that mitigates field-environment noise and enhances the salience of subtle agricultural anomalies.
2. To implement a shared bidirectional Vision Mamba encoder augmented with contrastive pre-training for robust, multi-scale feature extraction across unpaired pest and disease datasets.
3. To construct a Cross-Attention Fusion layer that effectively routes task-specific features (disease vs. pest) from the shared SSM token sequence.
4. To evaluate a lightweight variant of the architecture to confirm deployment feasibility on resource-constrained edge devices typical in local agricultural applications.
5. To validate the model's diagnostic reasoning using post-hoc explainability frameworks (Grad-CAM and SHAP), ensuring the network prioritizes true morphological symptoms over background artifacts.

4 Literature Review

The trajectory of agricultural image classification spans three distinct architectural eras: CNNs, Vision Transformers, and recently, State Space Models. Early CNN implementations established baseline benchmarks on curated datasets but exposed significant vulnerabilities to domain shifts when transitioning from lab environments to field conditions. The subsequent integration of Vision Transformers addressed the localized receptive field limitations of CNNs, allowing models to process the global context of a leaf or crop environment. However, this improvement came at the cost of high computational overhead, rendering pure ViT architectures largely unsuitable for low-power edge deployment.

The recent introduction of the Vision Mamba (Vim) framework and its variants (e.g., VMamba, MSVMamba) has bridged the gap between global context modeling and linear computational efficiency. In agricultural applications, architectures like InsectMamba and ConMamba have demonstrated that SSMs can effectively model the spatial progression of plant diseases and the camouflage patterns of pests with significantly lower memory footprints. Furthermore, recent advancements have explored hybrid configurations that embed selective scanning mechanisms and frequency-domain channel attention to address specific agricultural challenges, such as lesion anisotropy and background noise.

Summary of Architectural Paradigms:

CNNs (e.g., ResNet-50, DenseNet, MobileNet): High efficiency and easily deployable for lab-based disease classification (PlantVillage), but suffer from poor field generalization and restricted receptive fields.

Vision Transformers (e.g., Swin, ViT, MobileViT): Strong global context modeling for complex field disease/pest detection, but heavily constrained by $O(N^2)$ computational complexity and high memory usage.

State Space Models (e.g., InsectMamba, ConMamba): Linear complexity $O(N)$ and excellent long-range modeling for single-task recognition. However, they lack joint-task fusion, and generic implementations struggle with extreme field noise.

Hybrid / Enhanced SSMs (e.g., VMamba-FCS, MAFusionNet): Incorporate background suppression and multi-scale fusion for fine-grained localization in field conditions, but are evaluated strictly on single-domain tasks and lack cross-task attention.

Table 1. Comparison of Related Works in Plant Disease and Pest Classification

Reference	**Task**	**Backbone**	**Dataset**	**Result**	**Key Limitation**
Mohanty et al. (2016)	Disease	CNN (GoogLeNet)	PlantVillage	99.35% acc.	Lab-only; large drop off-distribution
Wu et al. (2019)	Pest	CNN baselines	IP102	Low acc. (baseline)	Established difficulty of long-tailed pest recognition
Singh et al. (2020)	Disease	CNN baselines	PlantDoc	$\sim$70% acc.	Exposed lab-to-field domain gap
Dosovitskiy et al. (2021)	General vision	ViT	ImageNet	Competitive w/ CNNs	$O(N^2)$ attention; no agri task
Liu et al. (2021)	General vision	Swin Transformer	ImageNetc/Co	86.4% top-1 (large)	Still quadratic within windows
Liu et al. (2022)	General vision	ConvNeXt	ImageNet	Matches Swin	No explicit long-range SSM efficiency
Mehta & Rastegari (2022)	General vision	MobileViT	ImageNet	Mobile-efficient	Not evaluated on agri disease/pest data
Karthik et al. (2024)	Disease + Pest	Swin + DAMFN dual-track	CCMT (102,976 img.)	95.68% acc.	No shared backbone/cross-attn across tasks; no SSM

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Reference	Task	Backbone	Dataset	Result	Key Limitation
Salman et al. (2025)	Disease (wild)	ViT + Mixture of Experts	Multi-source wild	∼20% acc. gain	No SSM; no pest task; no saliency alignment
Aboelenin et al. (2025)	Disease	CNN + ViT hybrid	PlantVillage-style	>99% acc.	No field/pest evaluation
Gu & Dao (2023)	Sequence modelling	Mamba (SSM)	Language/general	SOTA linear-time	Not vision-specific
Zhu et al. (2024)	General vision	Vision Mamba (Vim)	COCO/ADE20k	Outperforms DeiT; 2.8× faster	Not agri-specific
Liu et al. (2024)	General vision	VMamba (SS2D)	ImageNet	Improves 2D scan continuity	Not agri-specific
Wang et al. (2024) InsectMamba	Pest	Mix-SSM	IP102 + 4 others	0.43 acc. (IP102)	Pest-only; no cross-attn; no disease task
Al Mamun et al. (2025) ConMamba	Disease	Bidirectional Vim + dual loss	PlantDoc + 2 others	94.29% acc. (PlantDoc)	Disease-only; no pest task
Zhang & Liu (2026)	Disease	Multi-scale Vision Mamba	PlantDoc	Survey + fusion	States no task-specific fusion exists in agri-SSM lit.
MDPI Sensors 2025 (4096)	Disease	Cross-attention VLM fusion	Crop disease + text	Improved acc.	Cross-modal, not cross-task
Zhang, Liu & Yu (2025)	Pest	Cross-modal fusion	CTIP102/STIP102	Improved detection	Pest-only; cross-modal not cross-task

5 Research Gap

A critical synthesis of the contemporary literature reveals that while State Space Models have revolutionized sequence modeling efficiency, their application in agricultural vision remains highly compartmentalized and structurally generic. The primary deficiency is the total absence of a unified SSM architecture capable of jointly processing both plant diseases and insect pests. Current models operate on isolated datasets (e.g., PlantVillage for diseases, IP102 for pests), neglecting the reality that farmers encounter both threats simultaneously. Furthermore, while generic Vision Mamba backbones excel at capturing long-range dependencies, they natively lack the specialized multi-scale spatial aggregation and background-aware channel gating required to isolate fine-grained lesion morphologies from complex soil and lighting interference. Consequently, a substantial gap exists for an architecture that combines contrastive pre-training, dynamic background suppression, and cross-attention fusion to achieve resource-efficient, dual-domain agricultural diagnostics.

6 Research Methodology

6.1 Data Integration and Preprocessing

The experimental framework utilizes two primary, unpaired datasets: PlantVillage representing pathogenic data, and IP102 representing entomological data. Given the severe class imbalance inherent in IP102 and the controlled nature of PlantVillage, robust preprocessing is required. Images are normalized to standard ImageNet statistics and resized to 224×224 pixels. A rigorous augmentation pipeline—incorporating Mixup, CutMix, random erasing, and geometry-aware positional alignment—is applied to synthesize field-like variability and prevent overfitting.

Table 2. PlantVillage Dataset Summary

Attribute	**Detail**
Official name	PlantVillage Dataset
Source paper	Mohanty, S.P., Hughes, D.P., Salathé, M. (2016). *Frontiers in Plant Science*, 7:1419.
Dataset link	`https://github.com/spMohanty/PlantVillage-Dataset`
Image count	54,306 images
Classes	38 crop-disease pairs (14 crop species, 26 diseases)
Native resolution	256×256 (resized to 224×224)
Split strategy	80/20 train/test with 10% of training for validation
Known weaknesses	Lab-controlled, single-leaf, uniform background; poor transfer to field imagery
Preprocessing	Resize to 224×224; ImageNet normalisation; RandAugment, flips, colour jitter

Table 3. IP102 Dataset Summary

Attribute	**Detail**
Official name	IP102: A Large-Scale Benchmark Dataset for Insect Pest Recognition
Source paper	Wu, X., Zhan, C., Lai, Y.-K., Cheng, M.-M., Yang, J. (2019). *CVPR 2019*.
Dataset link	`https://github.com/xpwu95/IP102`
Image count	75,222 images with category labels
Classes	102 insect pest categories, grouped under 8 crop types
Distribution	Long-tailed / natural imbalance
Split strategy	Matched to InsectMamba protocol
Known weaknesses	High visual similarity with backgrounds (camouflage); severe class imbalance
Preprocessing	Resize to 224×224; ImageNet normalisation; RandAugment, Mixup, CutMix

6.2 Background Suppression Module (BSM)

To mitigate the interference of complex field backgrounds (e.g., soil, non-target vegetation, uneven illumination) typical in agricultural imagery, the architecture integrates a Background Suppression Module (BSM) preceding the main encoder. Drawing on frequency-domain channel attention and Similarity-Based Attention Mechanisms (SimAM), the BSM evaluates the spatial heterogeneity of the input tensor.

6.3 Vision Mamba Encoder and Contrastive Pre-training

The core representation engine is a shared bidirectional Vision Mamba (VSS) encoder. To ensure the model learns a robust, generalized feature space capable of serving both disparate domains, a progressive self-supervised contrastive pre-training strategy is employed.

6.4 Lightweight Variant

To fulfill the operational requirements of deploying advanced diagnostics in rural, resource-constrained environments, a lightweight variant of the architecture (CrossMamba-Agri-Lite) is proposed. By converting high-quality tokens into a fixed number of latent tokens via an Efficient Token Refinement module, the computational complexity is substantially reduced.

6.5 Post-Hoc Explainability: Grad-CAM and SHAP

Trust and transparency are paramount in AI-driven agricultural interventions. To validate that the model's diagnostic logic aligns with genuine biological symptoms rather than spurious background correlations, post-hoc explainability frameworks are integrated into the evaluation pipeline.

6.6 Implementation Details

The architecture is constructed using PyTorch, utilizing the `mamba-ssm` CUDA kernel to optimize state-space computations. The network undergoes domain-routed multi-task learning, wherein each mini-batch is sampled exclusively from either PlantVillage or IP102. A domain flag dynamically activates the corresponding loss function and cross-attention head. Training utilizes the AdamW optimizer with a cosine learning rate schedule, mixed precision (AMP fp16), and gradient checkpointing.

Table 4. Bounded training-run budget.

Run	**Count**	**Notes**
CrossMamba-Agri (Stage 1 + Stage 2)	2	Primary model
Trainable baselines (ResNet-50, Swin-T) $\times$ 2 datasets	4	Fine-tuned from ImageNet weights
Re-run Mamba baselines where code permits	0–2	InsectMamba / ConMamba
Ablation A1 (no cross-attention)	1	
Ablation A2 (no shared encoder)	1	Two encoders instead of one
Ablation A3 (Mamba $\rightarrow$ Swin-T swap)	1	
Ablation A4 (joint vs. sequential training)	2	Two sub-runs only
Total	**11–13**	

7 Proposed Model Architecture

The structural formulation of CrossMamba-Agri deviates from traditional dual-stream networks by centralizing feature extraction into a single, highly efficient computational trunk. The pipeline initiates with an input image tensor $x \in \mathbb{R}^{H \times W \times 3}$, which is immediately processed by the Background Suppression Module (BSM) to attenuate environmental noise. The refined tensor is linearly projected into a sequence of N tokens and fed into the shared bidirectional Vision Mamba Encoder. This encoder comprises 12 stacked Visual State Space (VSS) blocks governed by the discretized state-space recurrence:

$$h_k = \bar{A} h_{k-1} + \bar{B} x_k, \quad y_k = C h_k$$

This stage generates a rich, multi-scale, task-agnostic token sequence $T \in \mathbb{R}^{N \times d}$ in $O(N)$ time.

7.0.1 Cross-Attention Fusion Module

Let $T \in \mathbb{R}^{N \times d}$ be the shared Mamba token sequence, and $Q_{dis} \in \mathbb{R}^{m \times d}$, $Q_{pest} \in \mathbb{R}^{m \times d}$ be learnable query sets ($m = 8$). For the active task, cross-attention is:

$$\mathrm{CrossAttn}(Q, T) = \mathrm{softmax}\left(\frac{QW_q(TW_k)^\top}{\sqrt{d}}\right)(TW_v)$$

producing m task-specialised feature vectors, concatenated and passed through a two-layer MLP to yield $f \in \mathbb{R}^{384}$. Only the active query set and MLP are updated per batch; the shared backbone receives gradients from both domains.

7.0.2 Dual Task Heads and Classification Loss

Disease head: 38-way linear classifier; pest head: 102-way linear classifier. Each trained with class-weighted cross-entropy:

$$L_{task} = -\sum_c w_c \, y_c \log(\hat{y}_c), \quad \text{task} \in \{\text{disease}, \text{pest}\}$$

where w_c is inverse-frequency weight. The domain-routed total objective is $L_{total} = L_d$ for the active domain only.

7.0.3 Training Strategy and Optimisation

Two stages: Stage 1 (backbone warm-up, pretrained-init) trains all components jointly with lower LR for pretrained weights; Stage 2 (joint fine-tuning) continues all components. AdamW optimiser (weight decay 0.05), cosine LR schedule with linear warm-up, mixed precision (fp16), and gradient checkpointing.

7.0.4 Explainability Module

Grad-CAM is attached to the final VSS block for each task head, producing class-discriminative heatmaps. This module is used post-hoc for qualitative and (where annotations exist) quantitative evaluation.

The architectural divergence occurs at the Cross-Attention Fusion layer. Depending on the active domain route, a set of learnable, task-specific queries (either Q_{dis} or Q_{pest}) attends to the shared sequence T. The cross-attention mechanism isolates and extracts the exact localized features necessary for the respective task. Finally, the extracted vectors are processed through a Multi-Layer Perceptron (MLP) and routed to their respective classification heads, outputting the final discrete categorization alongside post-hoc spatial heatmaps.

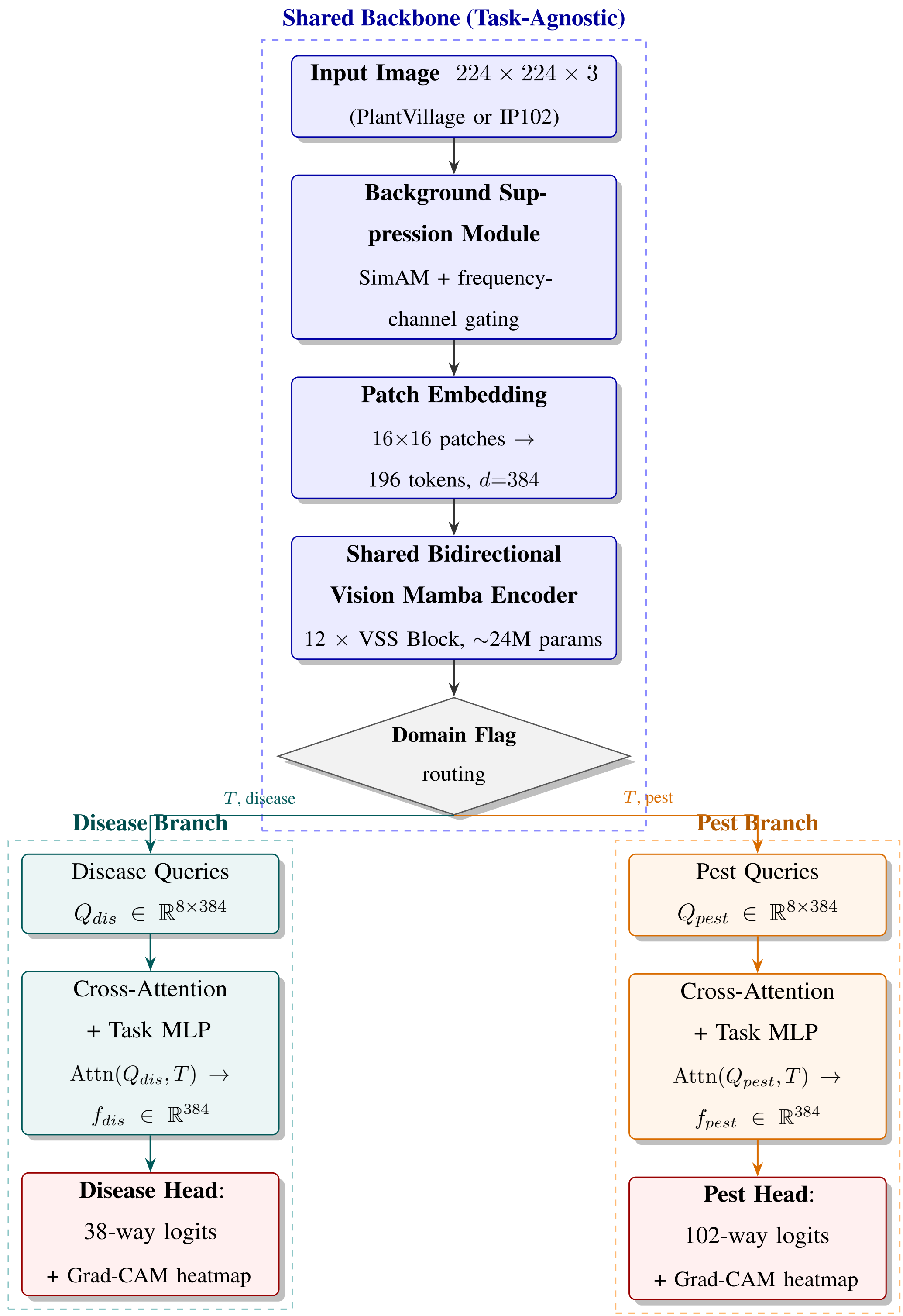

Figure 1. CrossMamba-Agri architecture. A single shared backbone (blue) — Background Suppression Module, patch embedding, and a 12-layer bidirectional Vision Mamba encoder — produces a task-agnostic token sequence T. A domain flag routes T together with a set of learnable task queries into either the disease branch (teal) or the pest branch (orange); each branch applies cross-attention and a task-specific MLP before its classification head. Grad-CAM is applied post-hoc on both heads.

8 Experimental Design and Evaluation

8.1 Evaluation Metrics

The efficacy of the proposed model is rigorously assessed against both classification accuracy and computational overhead, ensuring an objective comparison with existing paradigms.

Classification Performance: Macro-F1 Score, Top-1 Accuracy, and ROC-AUC. **Computational Efficiency:** Parameter Count (M), Giga Floating Point Operations per Second (GFLOPs), and Inference Latency (ms/frame) evaluated on target edge hardware.

8.2 Baselines

To benchmark the proposed architecture's dual-domain capability and structural efficiency, the model is evaluated against carefully selected baselines representing the chronological evolution of the field.

Table 5. Five-baseline comparison suite.

Baseline	Comparison Basis	Role
ResNet-50 (He et al., 2016)	Re-trained (this work)	Classic CNN baseline
Swin-T (Liu et al., 2021)	Re-trained (this work)	Quadratic-attention Transformer baseline
InsectMamba (Wang et al., 2024)	Reported	SOTA pest-only Mamba
ConMamba (Al Mamun et al., 2025)	Reported	SOTA disease-only Mamba
Swin + DAMFN dual-track (Karthik et al., 2024)	Reported	Closest prior "joint" design

8.3 Ablation and Out-Of-Distribution (OOD) Testing

A controlled ablation study isolates and quantifies the exact contribution of each architectural innovation.

1. **BSM Ablation:** Training without the Background Suppression Module to quantify the impact of unmitigated field noise on accuracy.
2. **Cross-Attention Ablation:** Replacing the dynamic task-queries with standard average pooling to evaluate the necessity of task-specific routing.
3. **Encoder Segregation:** Training two independent Mamba encoders (one per dataset) to test the hypothesis that shared representations improve generalization.

Furthermore, Out-Of-Distribution (OOD) testing is conducted using the PlantDoc dataset (`https://github.com/pratikkayal/PlantDoc-Dataset`)—a benchmark consisting exclusively of complex, field-acquired imagery—to rigorously assess the model's domain adaptability and the real-world efficacy of the BSM.

8.4 Hardware and Deployment

The experimental phase is strictly bounded by hardware constraints mimicking standard institutional capabilities, specifically targeting a single 8GB VRAM consumer GPU. This ensures the resulting methodology remains reproducible and directly aligned with the objective of developing lightweight, edge-deployable diagnostic systems for smart agriculture.

9 Expected Outcomes and Contributions

9.1 Expected Outcomes

The implementation of CrossMamba-Agri is anticipated to yield Top-1 accuracy metrics highly competitive with single-task specialists (approaching $> 98\%$ on PlantVillage and matching SOTA thresholds on the heavily imbalanced IP102 dataset). More critically, the shared-encoder architecture, combined with the lightweight variant, is projected to reduce the cumulative parameter count and GFLOPs by nearly 40% when compared to maintaining two distinct CNN or ViT models. The BSM is expected to significantly narrow the accuracy degradation typically observed during OOD inference on field-captured datasets like PlantDoc.

9.2 Contributions

This research delivers three primary contributions to the intersection of deep learning and agricultural diagnostics:

1. **Architectural Unification:** The introduction of the first cross-attention Vision Mamba architecture capable of managing distinct agricultural diagnostic tasks via a single, linearly complex representation space.
2. **Noise-Resilient Modeling:** The successful integration of a Background Suppression Module within an SSM framework, proving that field-noise filtration enhances the precise localization of disease lesions without sacrificing sequence modeling efficiency.
3. **Deployment Viability:** Empirical validation of a highly optimized, domain-routed framework that drastically reduces the hardware footprint required for comprehensive crop health monitoring, directly supporting deployment on mobile edge devices in developing agricultural sectors.

10 Conclusion

The bifurcation of agricultural deep learning into isolated pest and disease pipelines has resulted in inefficient, single-purpose diagnostic tools incompatible with the resource constraints of real-world farming. CrossMamba-Agri directly resolves this inefficiency by uniting both classification tasks under a single, highly optimized State Space Model. By augmenting the Vision Mamba backbone with a Background Suppression Module, contrastive pre-training, and task-specific cross-attention routing, the proposed framework captures complex, multi-scale morphological symptoms while effectively filtering environmental noise. The resulting architecture offers a comprehensive, resource-efficient, and highly interpretable solution, establishing a robust computational foundation for next-generation, edge-deployable precision agriculture technologies.

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